

Goal-Based Investing under Ergodic Dynamics

Time-Average Growth, Geometric-Mean Targets, and a Fixed-Point Closure of the Constraint Identity

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We reformulate the goal-based portfolio problem under geometric Brownian motion with the time-average growth rate of log wealth as the objective and a geometric-mean terminal target as the constraint. The reformulation produces an exact fixed-point closure of the constraint identity: the contribution stream's effect on its own discount base yields a future-value-of-annuity calculation in which the time-average growth rate $R(l) = \mu_b + \mu_e l - \frac{1}{2}\sigma^2 l^2$ replaces the ensemble drift. The optimal allocation is Kelly. The framework presents a one-parameter frontier of (l, c) trade-offs that the investor traverses by preference between volatility and contribution dollars. Model-predictive re-planning from the realized state recovers the dynamic-programming optimum exactly, because the constraint is linear in $\ln P$ and the objective integrand is stationary.

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Introduction

The goal-based portfolio problem asks how an investor with current wealth $P(t)$, terminal target P_T , horizon T , and a continuous contribution stream $c(t)$ should allocate between a risky and a risk-free asset. The standard formulation, following the goal-based wealth management literature (Brunel, 2015; Das, Ostrov, Radhakrishnan, and Srivastav, 2020), poses this as an expected-utility maximization subject to an expected terminal wealth constraint $\mathbb{E}[P(T)] = P_T$.

Both objects are ensemble averages. Under multiplicative wealth dynamics, the ensemble drift $\mu_b + \mu_e l$ governs $\mathbb{E}[P(t)]$, while the time-average growth rate of a single path is $R(l) = \mu_b + \mu_e l - \frac{1}{2}\sigma^2 l^2$. The gap between them is the volatility drag $\frac{1}{2}\sigma^2 l^2$, the difference between the average outcome across paths and the typical outcome along a path (Peters and Adamou, 2018; Peters, 2019). For an investor who walks one path, the second number is the relevant one.

This note reformulates the problem in terms of objects that align with the investor's experience. The objective becomes the expected integral of $d \ln P$. The constraint becomes a geometric-mean target, $\mathbb{E}[\ln P(T)] = \ln P_T$. Both are now phrased in $\ln P$, which is the ergodic transformation for geometric Brownian motion: the change of variables under which time and ensemble averages of the increment process coincide.

The reformulation has four consequences.

First, the constraint identity admits an exact fixed-point closure. Contributions feed back through the wealth base they build, and the resulting expression for required contributions takes the form of a standard future-value-of-annuity calculation, with the time-average growth rate $R(l)$ replacing the ensemble drift.

Second, the optimal allocation is Kelly, $l^* = \mu_e/\sigma^2$, with the savings rate as the residual control. The framework leaves the allocation pinned at Kelly unless the investor's preferences over volatility pull it away.

Third, the framework presents the investor a one-parameter frontier of (l, c) trade-offs at each rebalancing date. Deviations from Kelly are admissible below l^* as volatility-versus-savings trade-offs and dominated above.

Fourth, the model-predictive re-planning structure that arises naturally from this construction recovers the dynamic-programming optimum exactly, because the constraint is linear in $\ln P$ and the integrand of the objective is stationary.

The paper restricts attention to constant-parameter geometric Brownian motion, where the ergodic transformation is the natural log and the closed forms are elementary. Section sets up the dynamics and states the problem. Section derives the constraint identity and its fixed-point closure. Sections and present the two user-facing frontiers, snapshot and lifecycle, and characterize the Kelly minimum. Section develops the feasibility window under contribution and allocation bounds. Section states the model-predictive algorithm and its equivalence to dynamic programming. Section discusses extensions. Derivations and proofs are in the appendix.

Setup

The portfolio value $P(t)$ evolves under

$$dP(t) = [P(t)(\mu_b + \mu_e l(t)) + c(t)] dt + P(t) \sigma l(t) dW_t, \quad P(0) = P_0, \quad (1)$$

where W is a standard Brownian motion, $l(t)$ is the risky-asset allocation, and $c(t) \geq 0$ is the contribution rate.

We assume throughout that the risk-free rate μ_b , excess return μ_e , and volatility σ are positive constants, and that the horizon $T < \infty$ and target $P_T > 0$ are fixed.

Define the time-average log-growth rate at allocation l :

$$R(l) \equiv \mu_b + \mu_e l - \frac{1}{2} \sigma^2 l^2. \quad (2)$$

This is the deterministic drift of $\ln P$ in the absence of contributions, and the rate that governs the typical path of P .

The investor's problem is

$$\max_{l(\cdot), c(\cdot)} \mathbb{E} \left[\int_0^T d \ln P(s) \right] \quad \text{subject to} \quad \mathbb{E}[\ln P(T)] = \ln P_T \quad (3)$$

and the dynamics in (1). The objective and constraint are both phrased in $\ln P$, which is the ergodic transformation for GBM: $\ln P$ has stationary Gaussian increments, so time and ensemble averages of $d \ln P$ coincide along any single realized path.

The Constraint Identity and Its Closure

The constraint identity

Applying Itô's lemma to $\ln P$ under (1):

$$d \ln P = \left[R(l) + \frac{c}{P} \right] dt + \sigma l dW. \quad (4)$$

Integrate from t to T , take expectations conditional on $\mathcal{F}_t$, and impose $\mathbb{E}[\ln P(T)] = \ln P_T$:

$$\ln(P_T/P(t)) = \int_t^T R(l(s)) ds + \int_t^T \mathbb{E} \left[\frac{c(s)}{P(s)} \mid \mathcal{F}_t \right] ds. \quad (5)$$

The first term is the deterministic log-growth from the allocation policy. The second is the conditional expected log-growth contributed by savings. Equation (5) is the working identity of the framework.

The fixed-point closure

The second integrand in (5) cannot be evaluated directly because $P(s)$ in the denominator is the wealth process, which itself depends on contributions made between t and s . Define the cumulative contribution-induced log-growth

$$\phi(s) \equiv \int_t^s \overline{c(u)/P(u)} du, \quad (6)$$

where $\overline{c/P}$ denotes the time-average expectation along the typical path. The self-reference in (6) closes into a separable ODE for ϕ , with solution

$$\phi(s) = \ln \left[1 + \frac{c}{P(t)} A(R(l), s - t) \right], \quad A(R, \tau) \equiv \frac{1 - e^{-R\tau}}{R}, \quad (7)$$

when l and c are held constant on $[t, T]$. The function $A(R, \tau)$ is the standard continuous-time annuity factor at rate R over horizon τ . The derivation is in Appendix .

Substituting $\phi(T)$ into (5), exponentiating, and rearranging produces the working form of the constraint, which we call the *dollar-balance equation*:

$$\boxed{P(t) e^{R(l)(T-t)} + c \cdot \frac{e^{R(l)(T-t)} - 1}{R(l)} = P_T.} \quad (8)$$

This is the standard future-value-of-annuity identity from finance: current wealth $P(t)$ compounded forward at rate $R(l)$, plus a continuous-flow annuity c compounded forward at the same rate, equals the terminal target P_T . The rate $R(l)$ is the time-average growth rate of log wealth, not the ensemble drift $\mu_b + \mu_e l$, so the volatility drag is built into the compounding by construction. The derivation is in Appendix .

The textbook future-value-of-annuity formula uses $\mu_b + \mu_e l$ as the rate. The replacement of this rate by $R(l)$ is the working content of the reformulation. Numerically, for $\sigma = 0.18$ and $l = 1$, the difference is $\frac{1}{2}\sigma^2 l^2 = 1.62\%$ per year, which compounds to a thirty-percent gap over a twenty-year horizon.

The Snapshot Frontier

Solving (8) for c in terms of l gives the snapshot frontier:

$$\boxed{c(l) = \frac{P_T - P(t) e^{R(l)(T-t)}}{(e^{R(l)(T-t)} - 1)/R(l)}.} \quad (9)$$

The frontier is the set of (l, c) pairs that, held constant from t to T , satisfy the geometric-mean terminal constraint.

Whenever the goal exceeds the Kelly-compounded current wealth (that is, $P_T > P(t) e^{R(l^*)(T-t)}$ with $l^* = \mu_e/\sigma^2$), $c(l)$ is strictly convex on $[l_{\min}, l_{\max}]$ with global minimum at the Kelly allocation l^* . The Kelly allocation maximizes $R(l)$ over $\mathbb{R}$, which minimizes the compounded shortfall in the numerator

and the annuity factor in the denominator; the former effect dominates. The derivation is in Appendix .

For $l > l^*$, both $c(l)$ and $\text{Var}(P(T))$ are increasing in l . Allocations above Kelly are therefore dominated under the joint criterion of minimizing required contributions and minimizing portfolio variance. The investor's admissible choice set is $l \in [l_{\min}, l^*]$. The trade-off along this segment is between allocation volatility and contribution level: lower l reduces portfolio variance at the cost of higher c .

The Lifecycle Frontier

A more realistic specification lets contributions grow with income. Set

$$c(s) = c_0 e^{gs}, \quad (10)$$

with $c_0 > 0$ and $g \in \mathbb{R}$ deterministic. The same fixed-point argument applied to this deterministic contribution schedule produces the lifecycle constraint:

$$P(t) e^{R(l)(T-t)} + c_0 e^{gt} e^{R(l)(T-t)} \cdot \frac{e^{(g-R(l))(T-t)} - 1}{g - R(l)} = P_T. \quad (11)$$

The case $g = R(l)$ is handled by continuity: $(e^{(g-R(l))(T-t)} - 1)/(g - R(l)) \rightarrow T - t$. The derivation is in Appendix .

Solving for c_0 at fixed (g, l) :

$$c_0 = \frac{[P_T e^{-R(l)(T-t)} - P(t)] e^{-gt} (g - R(l))}{e^{(g-R(l))(T-t)} - 1}. \quad (12)$$

The investor specifies the wage-growth assumption g and the allocation l , and reads off the required initial contribution.

At fixed $g \neq R(l^*)$, c_0 is strictly convex in l with minimum at $l^* = \mu_e/\sigma^2$ on the interior of the feasibility region. The same dominance argument as for the snapshot frontier restricts the investor's admissible choice set to $l \in [l_{\min}, l^*]$. The derivation is in Appendix .

The Feasibility Window

The investor's allocation is bounded by risk-tolerance limits $l \in [l_{\min}, l_{\max}]$ and her contribution by dollar capacity $c \in [\text{LB}, \text{UB}]$. These bounds imply, at any state $(P(t), t)$, a window of reachable terminal wealth under the hold-to-expiry assumption.

Define $l_{\text{best}} = \min(l_{\max}, \mu_e/\sigma^2)$, and let $l_{\text{worst}} \in \{l_{\min}, l_{\max}\}$ denote the endpoint minimizing $R(l)$ over $[l_{\min}, l_{\max}]$. With constant l and constant c held to expiry, the set of reachable terminal wealth values is

$P_T \in [P(T)_{\min}, P(T)_{\max}]$, where

$$P(T)_{\max} = P(t) e^{R(l_{\text{best}})(T-t)} + \text{UB} \cdot \frac{e^{R(l_{\text{best}})(T-t)} - 1}{R(l_{\text{best}})}, \quad (13)$$

$$P(T)_{\min} = P(t) e^{R(l_{\text{worst}})(T-t)} + \text{LB} \cdot \frac{e^{R(l_{\text{worst}})(T-t)} - 1}{R(l_{\text{worst}})}. \quad (14)$$

The window expands when $P(t)$ rises and contracts when it falls, each rebalancing period. A goal $P_T > P(T)_{\max}$ is infeasible from the current state and signals a required revision of P_T , T , or the bounds.

Model-Predictive Algorithm and Bellman Equivalence

Algorithm

At each rebalancing time $t \in [0, T)$:

1. Observe the realized portfolio value $P(t)$.
2. Compute the feasibility window $[P(T)_{\min}, P(T)_{\max}]$ via (13) and (14).
3. If $P_T \notin [P(T)_{\min}, P(T)_{\max}]$, prompt for revision of P_T , T , or bounds.
4. Otherwise, compute the chosen frontier: (9) (snapshot) or (12) (lifecycle).
5. The investor selects (l, c) on the frontier subject to bounds.
6. Hold the policy over $[t, t + \Delta]$. Return to step 1.

Every step is closed-form. The control rule itself involves no approximation of $1/P(s)$.

Bellman equivalence

Let $J^*(P(t), t)$ denote the optimal value of (3) under the geometric-mean constraint from initial state $(P(t), t)$. The policy obtained by re-solving (5) at each t from the realized $P(t)$ coincides almost surely with the optimal feedback control of the dynamic-programming problem with value function J^* . The argument turns on two properties of the reformulated problem: the integrand of the objective is stationary in $(l, c/P)$, and the constraint is linear in $\ln P$. Together, these properties make the re-solved problem from $(P(t), t)$ the dynamic-programming subproblem of the original. The derivation is in Appendix .

Under the arithmetic-mean constraint $\mathbb{E}[P(T)] = P_T$, the analogous equivalence fails: the variational re-resolution is on $\mathbb{E}[P]$ but the HJB optimum is on $\mathbb{E}[\ln P]$, and the Jensen gap is exactly the volatility drag absorbed by $R(l)$ in the geometric-mean formulation.

Extensions

Multiple risky assets. For an n -vector of risky assets with excess return $\mu_e \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{R}^{n \times n}$, the Kelly allocation generalizes to $l^* = \Sigma^{-1} \mu_e$, with $R(l) = \mu_b + \mu_e^\top l - \frac{1}{2} l^\top \Sigma l$. The dollar-balance form (8) and the fixed-point closure are unchanged.

Time-varying parameters. Piecewise-constant (μ_b, μ_e, σ) extend the framework by recomputing the constraint identity each period. The closed forms hold within each segment.

Merton recovery. Setting $P_T = P_0 \exp(rT)$ for $r = \mu_b + \mu_e^2 / (2\sigma^2)$ (the Kelly-compounded current wealth) gives $c = 0$ at the Kelly allocation; the framework reduces to the unconstrained log-optimal portfolio.

Anomalous diffusions. The natural log is the ergodic transformation specific to geometric Brownian motion. For subdiffusive, superdiffusive, or heavy-tailed processes, the corresponding generalized log replaces $\ln$ throughout, and the fixed-point closure carries over with the appropriate annuity factor. A companion note develops this generalization.

Conclusion

The reformulation of the goal-based portfolio problem around the time-average growth rate of log wealth and a geometric-mean terminal target produces a framework with three operational properties: an exact fixed-point closure of the constraint identity, a closed-form (l, c) frontier with a Kelly minimum, and a model-predictive re-planning loop that recovers the dynamic-programming optimum exactly. The framework respects the experience of an investor on a single realized path rather than the expectation across a population of paths she does not traverse. The implementation is elementary at each rebalancing date, and the same construction extends to multiple assets, time-varying parameters, and anomalous diffusions through the appropriate generalized log.

Technical Appendix

Derivation of the fixed-point closure

Let $m(s) = \mathbb{E}[\ln P(s) \mid \mathcal{F}_t]$. Taking conditional expectations of (4):

$$m'(s) = R(l) + \overline{c/P(s)}. \quad (15)$$

Define $\overline{1/P(s)} \equiv 1/\exp(m(s))$, the ergodic expectation of $1/P(s)$ under the time-average framing. Since the contribution $c(s) = c$ is deterministic,

$$\overline{c/P(s)} = c \cdot \overline{1/P(s)} = \frac{c}{e^{m(s)}}. \quad (16)$$

Therefore

$$m'(s) = R(l) + c e^{-m(s)}, \quad (17)$$

with $m(t) = \ln P(t)$. Substituting $\phi(s) = m(s) - \ln P(t) - R(l)(s - t)$ so that $\phi(t) = 0$ and $m(s) = \ln P(t) + R(l)(s - t) + \phi(s)$:

$$\phi'(s) = c e^{-\ln P(t) - R(l)(s-t) - \phi(s)} = \frac{c}{P(t)} e^{-R(l)(s-t)} e^{-\phi(s)}. \quad (18)$$

This is a separable ODE. Multiply both sides by $e^{\phi(s)}$:

$$e^{\phi(s)} \phi'(s) = \frac{c}{P(t)} e^{-R(l)(s-t)}. \quad (19)$$

The left side equals $\frac{d}{ds} [e^{\phi(s)}]$. Integrating from t to s with $e^{\phi(t)} = 1$:

$$e^{\phi(s)} - 1 = \frac{c}{P(t)} \int_t^s e^{-R(l)(u-t)} du = \frac{c}{P(t)} A(R(l), s - t). \quad (20)$$

Solving for $\phi(s)$ gives (7).

Derivation of the dollar-balance equation

Substituting $\phi(T)$ from (7) into the constraint identity (5) with constant l :

$$\ln(P_T/P(t)) = R(l)(T - t) + \ln \left[1 + \frac{c}{P(t)} A(R(l), T - t) \right]. \quad (21)$$

Exponentiating:

$$\frac{P_T}{P(t)} e^{-R(l)(T-t)} = 1 + \frac{c}{P(t)} A(R(l), T - t). \quad (22)$$

Multiplying both sides by $P(t) e^{R(l)(T-t)}$:

$$P_T = P(t) e^{R(l)(T-t)} + c e^{R(l)(T-t)} A(R(l), T - t). \quad (23)$$

Using $e^{R\tau} A(R, \tau) = (e^{R\tau} - 1)/R$ gives (8).

Convexity of the snapshot frontier

Write $\tau = T - t$ and $S(l) = (e^{R(l)\tau} - 1)/R(l)$ for the future-value annuity factor. From (9),

$$c(l) = \frac{P_T - P(t) e^{R(l)\tau}}{S(l)}. \quad (24)$$

Differentiating with respect to l and using $R'(l) = \mu_e - \sigma^2 l$:

$$c'(l) = -\frac{R'(l)}{S(l)^2} \left[P(t) e^{R(l)\tau} \tau S(l) + (P_T - P(t) e^{R(l)\tau}) S'(l)/R'(l) \cdot R'(l) \right]. \quad (25)$$

A cleaner route is to differentiate the implicit form. From the dollar balance (8),

$$P(t) e^{R(l)\tau} + c(l) S(l) = P_T. \quad (26)$$

Differentiating:

$$P(t) \tau e^{R(l)\tau} R'(l) + c'(l) S(l) + c(l) S'(l) = 0, \quad (27)$$

so

$$c'(l) = -\frac{P(t) \tau e^{R(l)\tau} R'(l) + c(l) S'(l)}{S(l)}. \quad (28)$$

At $l^* = \mu_e/\sigma^2$, $R'(l^*) = 0$ and $S'(l^*) = 0$ (the latter because S depends on l only through $R(l)$, and $R'(l^*) = 0$). Therefore $c'(l^*) = 0$. The second-order condition reduces, after similar substitution and using $R''(l) = -\sigma^2$, to

$$c''(l^*) = -\frac{P(t) \tau e^{R(l^*)\tau} R''(l^*)}{S(l^*)} = \frac{P(t) \tau e^{R(l^*)\tau} \sigma^2}{S(l^*)} > 0 \quad (29)$$

under the stated condition $P_T > P(t) e^{R(l^*)\tau}$ (so $c(l^*) > 0$ and $S(l^*) > 0$). Strict convexity follows by continuity and the absence of other critical points on $[l_{\min}, l_{\max}]$ since $R'(l)$ vanishes only at l^* .

Derivation of the lifecycle constraint

The fixed-point argument of Appendix extends to deterministic $c(s) = c_0 e^{gs}$. The ODE for ϕ becomes

$$\phi'(s) = \frac{c_0 e^{gs}}{P(t)} e^{-R(l)(s-t)} e^{-\phi(s)}. \quad (30)$$

Multiplying by $e^{\phi(s)}$ and integrating:

$$e^{\phi(s)} - 1 = \frac{c_0}{P(t)} \int_t^s e^{(g-R(l))(u-t)+R(l)(0)} du, \quad (31)$$

which simplifies to

$$e^{\phi(s)} - 1 = \frac{c_0 e^{gt}}{P(t)} \cdot \frac{e^{(g-R(l))(s-t)} - 1}{g - R(l)}. \quad (32)$$

At $s = T$, substituting $\phi(T)$ into (5), exponentiating, and multiplying through by $P(t) e^{R(l)(T-t)}$ yields (11).

Convexity of the lifecycle frontier

Differentiating (12) implicitly through the dollar balance, the same structure as Appendix applies: c_0 is convex in l because the dominant l -dependence enters through $R(l)$, and R is concave with maximum at Kelly. The implicit-function argument carries over with $R'(l^*) = 0$ giving $c'_0(l^*) = 0$ and $R''(l^*) = -\sigma^2$ giving $c''_0(l^*) > 0$ on the interior of feasibility.

Bellman equivalence under the geometric-mean constraint

The dynamic-programming subproblem from state $(P(t), t)$ is

$$J^*(P(t), t) = \sup_{l, c} \mathbb{E} \left[\int_t^T d \ln P(s) \mid \mathcal{F}_t \right] \quad \text{s.t.} \quad \mathbb{E}[\ln P(T) \mid \mathcal{F}_t] = \ln P_t. \quad (33)$$

Two observations close the proof.

First, the objective integrand $d \ln P/dt = R(l) + c/P$ depends on the controls (l, c) and the state P only through stationary functions: $R(l)$ depends on l alone, and c/P on the contemporaneous control and state. There is no explicit time dependence and no path dependence through $\ln P$ history.

Second, the constraint is linear in $\ln P(T)$, with $\ln P(t)$ entering only as a shift: $\mathbb{E}[\ln P(T) \mid \mathcal{F}_t] - \ln P(t) = \ln(P_T/P(t))$, which depends on the realized state $P(t)$ but not on the path leading to it.

Together, these properties imply that the constraint identity (5) rewritten from $(P(t), t)$ has the same functional form as from $(P_0, 0)$, with $P(t)$ in place of P_0 and $T-t$ as the remaining horizon. The re-solved policy is therefore the optimal continuation by the principle of optimality (Bellman, 1957).

Further Reading and References

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